

SPRING 2020 MATH 4620/6620 Numerical Analysis II

Instructor

Dr. Xiaojing Ye (E-mail: xye@gsu.edu).

Course Website

<https://math.gsu.edu/xye/course/na2.html>

Office Hours and Location

M 10:30–11:30 AM, W 9:30–10:30 AM, 25 Park Place, Room 1436.

Class Time and Location

MW 2:00–3:15 PM, Classroom South 101.

Textbook

Numerical Analysis (10th edition), Burden and Faires, Brooks/Cole (Cengage Learning).

Prerequisites

Grade of C or higher in Math 2215 Multivariate Calculus or equivalent and the ability to program in a high-level language (e.g., MATLAB and Python).

Course Topics

Numerical techniques for ordinary differential equations: implicit methods and predictor-corrector schemes, Runge-Kutta methods, the Adams Families; Gaussian Elimination for linear systems, the LU factorization, stability, SPD matrices and Cholesky Decomposition, iterative methods for linear and nonlinear systems; Finite difference methods for partial differential equations.

Attendance and Administrative Drop Policy

Attendance to all lectures are required and checked during the semester. Students who do not the class regularly will be warned or even administratively dropped.

Homework

There are a total of 6 homework assignments (5 points each, 30 points total) during the semester, approximately one assignment every 1.5–2 weeks. Homework problems are taken from the exercises in the textbook, and the problem numbers and due time are announced during lectures.

- Homework are due on iCollege at <https://icollege.gsu.edu>.
- Homework submission instruction on <https://math.gsu.edu/xye/course/hwpolicy.html>.
- *Late homework and homework not following the instruction will not be accepted or graded.*

Exams

There are two midterm exams and one final exam. All exams are held in the lecture room.

- Exams are closed book/notes. Scratch papers are provided. Bring pen/pencil, Panther ID, and a basic calculator. *Smartphones are not allowed for any use throughout the test.*
- *Request for make-up exams must be approved by the instructor at least 10 days before the actual exam.* Unless in an extreme verifiable emergency, all make-ups must be worked out ahead of the time.
- Exam dates and time:
 - Midterm Exam 1 (20 points): 2:00–3:15 PM, Wednesday, February 12, 2020.
 - Midterm Exam 2 (20 points): 2:00–3:15 PM, Wednesday, March 25, 2020.
 - Final Exam (30 points): 1:30–3:30 PM, Wednesday, April 29, 2020.

Grading Policy

Homework (30 pts) + Midterms (40 pts) + Final exam (30 pts) = Total (100 pts).

Score	0–59	60–69	70–76	77–79	80–82	83–86	87–89	90–92	93–96	97–100
Grade	F	D	C	C+	B–	B	B+	A–	A	A+

Study Guide

Lecture slides are available for download in advance. Paying close attention and taking good notes during the class are of paramount importance (write notes on printed slides during lectures are recommended). After the class read the book, the slides/notes, and do as many of the homework problems as you can prior to the next class. Try to get the remaining problems explained in the next class or during the office hours. *You are responsible for all material covered in class if you missed any class.*

Academic Honesty

All work submitted by a student for a grade must have been done without help from other persons. *Cheating and plagiarism are strictly forbidden.* A first occurrence will result in a grade of 0 on the assignment *for all concerned parties* as well as an Academic Dishonesty form being filed with the Dean of Students. A second occurrence will result in a grade of F for the course for the concerned parties and a second Academic Dishonesty form being filed. See the University's policy on academic honesty at <http://codeofconduct.gsu.edu/>.

Withdrawal

Withdrawal on or before Midpoint day will result in a grade of "PW". Hardship withdrawal after the Midpoint day will result in a grade of "WF".

Inclement Weather Policy

If the University is closed due to inclement weather, any exam that may have been scheduled for that date will be administered on the next available class date.

Remarks

This course syllabus provides a general plan for the course; deviations may be necessary.